# Linear Functions & Systems Error Analysis TEACHER KEY



Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

## 1. Point-slope form: sign of $x_1$

**ORIGINAL PROBLEM**

Write the equation of the line through $(-3, 5)$ and $(2, -5)$ in slope-intercept form.

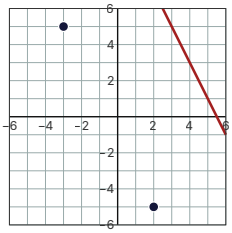
**INCORRECT WORKED SOLUTION**

$m = \frac{-5 - 5}{2 - (-3)} = \frac{-10}{5} = -2$

Use point $(-3, 5)$: $y - 5 = -2(x - 3)$

$y - 5 = -2x + 6$

$y = -2x + 11$



Dots: given points. Red: student's line.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The slope is right, but the point-slope form is wrong. With $x_1 = -3$, the factor is $x - (-3) = x + 3$, not $x - 3$. The graph shows the student's line misses both points; at $x = 2$ it gives $y = 7$, not $-5$.

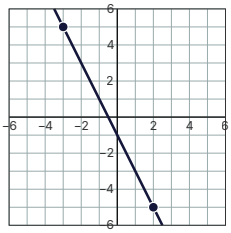
**B. SHOW CORRECT WORK (1 POINT)**

$y - 5 = -2(x - (-3)) = -2(x + 3)$

$y - 5 = -2x - 6$

$y = -2x - 1$

Check $(2, -5)$: $-2(2) - 1 = -5$



The line passes through both points.

## 2. Perpendicular slopes

**ORIGINAL PROBLEM**

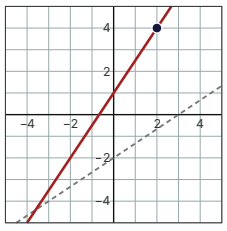
Line $\ell$ is $2x - 3y = 6$ (dashed). Write the equation of the line through $(2, 4)$ that is perpendicular to $\ell$.

**INCORRECT WORKED SOLUTION**

$y = \frac{2}{3}x - 2$, so the slope of $\ell$ is $\frac{2}{3}$.

Perpendicular slope: flip it, $m = \frac{3}{2}$.

$y - 4 = \frac{3}{2}(x - 2)$, so $y = \frac{3}{2}x + 1$



Dashed: $\ell$. Red: student's line.

**A. EXPLAIN THE MISTAKE (1 POINT)**

Perpendicular slopes are NEGATIVE reciprocals: flip the fraction AND change the sign, so the product is $-1$. The student only flipped it, so $\frac{2}{3} \cdot \frac{3}{2} = 1$. The red line is not perpendicular to $\ell$; it tilts the same way.

**B. SHOW CORRECT WORK (1 POINT)**

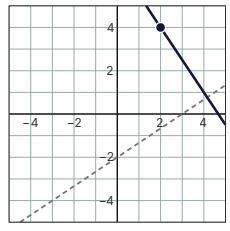
Perpendicular slope: $-\frac{3}{2}$,

since $\frac{2}{3} \cdot (-\frac{3}{2}) = -1$

$y - 4 = -\frac{3}{2}(x - 2)$

$y - 4 = -\frac{3}{2}x + 3$

$y = -\frac{3}{2}x + 7$



Dashed: $\ell$. Solid: the perpendicular line.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 3. Linear model: which variable is which?

**ORIGINAL PROBLEM**

A kayak rental costs $38 for 2 hours and $74 for 5 hours. The cost $C$ (dollars) is a linear function of the hours $h$. How many hours can you rent a kayak for $110?

**INCORRECT WORKED SOLUTION**

Rate: $\frac{74 - 38}{5 - 2}$ = $12 per hour, so $C = 12h + 14$.

For $110: $C = 12(110) + 14$

$C = 1320 + 14 = 1334$

You can rent the kayak for 1,334 hours.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The $110 is a COST, so it is an output ($C$), not an input ($h$). The student substituted dollars where hours belong, which gave an impossible 1,334 hours. The model is right; the substitution is wrong.

**B. SHOW CORRECT WORK (1 POINT)**

Cost is $110, so $C = 110$.

$12h + 14 = 110$

$12h = 96$, so $h = 8$ hours

Check: $12(8) + 14 = 110$

## 4. Solving a system by graphing

**ORIGINAL PROBLEM**

Solve the system by graphing: $y = 2x - 1$ and $x + y = 5$.

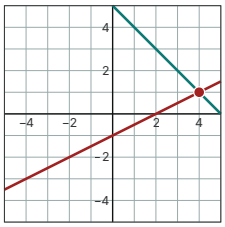
**INCORRECT WORKED SOLUTION**

Graph $y = 2x - 1$: start at $(0, -1)$. The slope is 2, so go right 2 and up 1.

Graph $x + y = 5$ through (0, 5) and (5, 0).

The lines cross at (4, 1).

Solution: (4, 1)




Red: student's $y = 2x - 1$. Teal: $x + y = 5$.

**A. EXPLAIN THE MISTAKE (1 POINT)**

Slope = $\frac{\text{rise}}{\text{run}}$, so a slope of 2 = $\frac{2}{1}$ means up 2 for every 1 to the right. The student went right 2 and up 1, which is slope $\frac{1}{2}$. Check (4, 1) in $y = 2x - 1$: $2(4) - 1 = 7$, not 1.

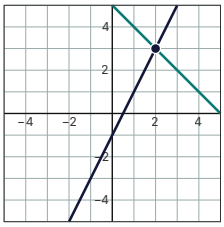
**B. SHOW CORRECT WORK (1 POINT)**

Slope 2 = $\frac{2}{1}$: from $(0, -1)$ go right 1 and up 2.

The lines cross at (2, 3).

Check: $2(2) - 1 = 3$ and $2 + 3 = 5$

Solution: (2, 3)




Navy: $y = 2x - 1$. Teal: $x + y = 5$.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 5. Dependent system

**ORIGINAL PROBLEM**

Solve the system: $2x - 4y = 6$ and $-3x + 6y = -9$.

**INCORRECT WORKED SOLUTION**

Multiply the first equation by 3 and the second by 2:

$6x - 12y = 18$ and $-6x + 12y = -18$

Add the equations: $0 = 0$

Both variables disappeared, so there is no solution.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The statement $0 = 0$ is always TRUE. When both variables cancel and the result is true, the two equations describe the same line, so there are infinitely many solutions. No solution is the case where the result is FALSE, such as $0 = 5$ (parallel lines).

**B. SHOW CORRECT WORK (1 POINT)**

$0 = 0$ is true for every $x$ and $y$.

Both equations simplify to $y = \frac{1}{2}x - \frac{3}{2}$, the same line.

Infinitely many solutions: every point on $2x - 4y = 6$.

Compare: a result like $0 = 5$ would mean no solution.

## 6. System of inequalities: shading

**ORIGINAL PROBLEM**

Graph the solution set of the system and shade it.

$y \geq -x + 1$

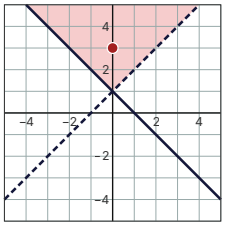
$y < x + 1$

**INCORRECT WORKED SOLUTION**

$y \geq -x + 1$: solid boundary, shade above.

$y < x + 1$: dashed boundary, shade above.

The solution is where the shading overlaps: the region above both lines.




Red region: student's. (0, 3) fails $y < x + 1$.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The boundary styles are right, but $y < x + 1$ means $y$-values LESS than the line, so shade BELOW the dashed line. Only ">" and "≥" shade above. Test: (0, 3) is in the student's region, but $3 < 0 + 1$ is false.

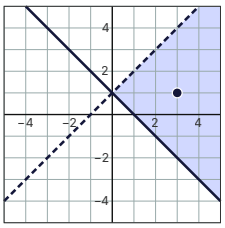
**B. SHOW CORRECT WORK (1 POINT)**

$y \geq -x + 1$: solid, shade above.

$y < x + 1$: dashed, shade BELOW.

The overlap is the wedge to the right of (0, 1).

Test (3, 1): $1 \geq -2$ and $1 < 4$, both true.




Solid line: included. Dashed line: not included.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 7. Dashed or solid boundary?

**ORIGINAL PROBLEM**

Graph the solution set of the system and shade it.

$y > 2x - 3$

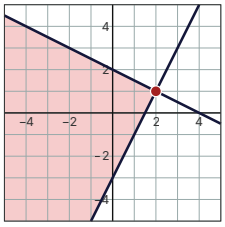
$y \leq -\frac{1}{2}x + 2$

**INCORRECT WORKED SOLUTION**

$y > 2x - 3$: solid boundary, shade above.

$y \leq -\frac{1}{2}x + 2$: solid boundary, shade below.

Shade the overlap. The corner (2, 1) is included.



Both solid. Red dot: (2, 1), wrongly included.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The shading is right, but a STRICT inequality (> or <) does not include its boundary, so $y > 2x - 3$ needs a DASHED line. Only ≤ and ≥ use solid lines. The corner (2, 1) lies on the dashed line: $1 > 2(2) - 3$ is $1 > 1$, which is false. The student's graph includes it.

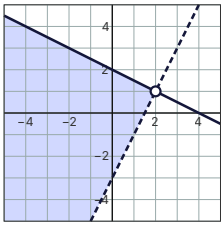
**B. SHOW CORRECT WORK (1 POINT)**

$y > 2x - 3$: DASHED line, shade above.

$y \leq -\frac{1}{2}x + 2$: solid line, shade below.

Test (2, 1): $1 > 1$ is false, so the corner is NOT included (open dot).

Test (0, 0): $0 > -3$ and $0 \leq 2$ are true, so (0, 0) is a solution.



Dashed: excluded. Solid: included. Open dot: (2, 1).

## 8. Line of best fit

**ORIGINAL PROBLEM**

Typing speed $y$ (words per minute) after $x$ weeks. Use regression to find the line of best fit, then predict the speed at 6.5 weeks.

$x$: 1, 2, 3, 4, 5, 6, 7, 8

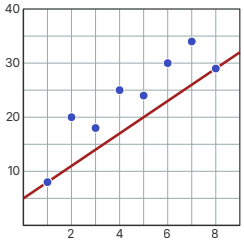
$y$: 8, 20, 18, 25, 24, 30, 34, 29

**INCORRECT WORKED SOLUTION**

Use the first and last points: (1, 8) and (8, 29).

$m = \frac{29 - 8}{8 - 1} = 3$, so $y = 3x + 5$

At $x = 6.5$: $y = 24.5$ words per minute



Red: student's line (first to last point).

**A. EXPLAIN THE MISTAKE (1 POINT)**

A line of best fit follows the overall trend of ALL the points, with data balanced above and below it. Using only the two end points ignores the rest: all 6 middle points lie above the student's line. Regression uses every point.

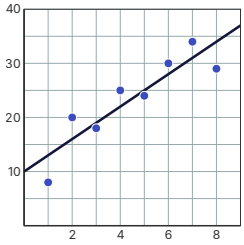
**B. SHOW CORRECT WORK (1 POINT)**

Regression on all 8 points gives $y = 3x + 10$.

Balanced: 4 points above the line, 4 below.

At $x = 6.5$: $y = 3(6.5) + 10$

$y = 29.5$ words per minute



Navy: line of best fit. $x$ = weeks, $y$ = words per minute.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 9. Linear programming: optimize at a vertex

**ORIGINAL PROBLEM**

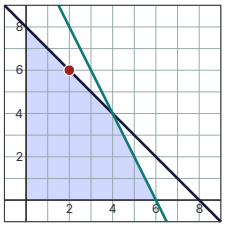
A bakery makes $x$ trays of muffins and $y$ trays of scones. Oven: $x + y \le 8$. Labor: $2x + y \le 12$. Also $x \ge 0$, $y \ge 0$. Profit: $P = 35x + 25y$. Find the maximum profit.

**INCORRECT WORKED SOLUTION**

The oven is the tight limit: pick a point on $x + y = 8$.

Try (2, 6): $P = 35(2) + 25(6) = 220$

Maximum profit: $220




Red dot: student's point (not a corner).

**A. EXPLAIN THE MISTAKE (1 POINT)**

The maximum of a linear objective occurs at a VERTEX (corner) of the feasible region. (2, 6) is in the middle of an edge, so it was never compared with the corners. The corner (4, 4) earns $240.

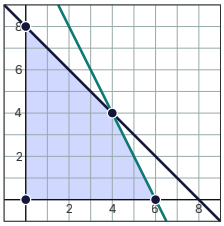
**B. SHOW CORRECT WORK (1 POINT)**

Vertices: (0, 0), (6, 0), (4, 4), (0, 8)

$P$ = 0, 210, 240, 200

Maximum profit: $240 at (4, 4)

Check (4, 4): $4 + 4 = 8$ and $2(4) + 4 = 12$




Dots: the four vertices of the feasible region.

## 10. Three variables: a lost term

**ORIGINAL PROBLEM**

Solve the system:

$x + y + z = 6$

$2x - y + z = 3$

$x + 2y - z = 2$

**INCORRECT WORKED SOLUTION**

Eliminate $z$ with Equation 1 − Equation 2:

$(x - 2x) + (y - y) + (z - z) = 6 - 3$

$-x = 3$, so $x = -3$

Equation 1 + Equation 3: $2x + 3y = 8$, so $y = \frac{14}{3}$

Then $z = 6 + 3 - \frac{14}{3} = \frac{13}{3}$

**A. EXPLAIN THE MISTAKE (1 POINT)**

The student lost the $y$ term: subtracting $-y$ gives $y - (-y) = 2y$, not 0. Subtracting a negative changes the sign. The correct new equation is $-x + 2y = 3$, and the messy fractions were a warning that something went wrong.

**B. SHOW CORRECT WORK (1 POINT)**

Eq. 1 − Eq. 2: $-x + 2y = 3$

Eq. 1 + Eq. 3: $2x + 3y = 8$

Solve the pair: $x = 1$, $y = 2$

$z = 6 - 1 - 2 = 3$. Solution: (1, 2, 3)